

Supplementary Material: Appendices A–O

A. BCa-only contrasts

These 14 of the 134 contrasts in the RECOMPUTED.md register described in Appendix I are the ones on which the primary (paired t) and the BCa bootstrap intervals disagree. Under the policy of §3.6 they carry no claim anywhere in this paper. In every one the direction of disagreement is the same, BCa excludes zero and t does not, so a reader who preferred BCa as primary would reach fourteen further positive verdicts. A t-non-significant flag here is not a claim that the effect is absent, and Table S1 lists the fourteen.

#	contrast	body	mean	paired t 95% CI	BCa 95% CI
1	prior – blank (held-out, ENDPOINT)	finger	–27.10	[–66.32, +12.12]	[–80.47, –3.85]
2	prior – blank (train, ENDPOINT)	finger	–34.29	[–70.93, +2.35]	[–72.79, –8.91]
3	prior – blank (held-out, EARLY)	finger	–5.88	[–16.15, +4.38]	[–20.25, –0.11]
4	coach cost – prior cost (ENDPOINT)	finger	+32.18	[–3.02, +67.39]	[+6.59, +68.56]
5	coach abrupt-withdrawn – blank (ENDPOINT)	elbow	–15.60	[–32.75, +1.54]	[–30.09, –0.87]
6	prior soft-withdrawn – blank (ENDPOINT)	finger	–29.49	[–67.41, +8.43]	[–78.26, –6.21]
7	prior soft-withdrawn – prior strict (ENDPOINT)	finger	–16.58	[–33.65, +0.49]	[–30.91, –1.86]
8	[priorcoach(const) – coach(const)] @EARLY – @ENDPOINT	elbow	–8.10	[–18.06, +1.87]	[–17.30, –0.22]
9	coach clean – coach contaminated (train, ENDPOINT)	finger	+37.71	[–1.62, +77.03]	[+4.90, +72.21]
10	priorcoach(const) – 0.5·[prior + coach(const)] (ENDPOINT)	finger	+9.50	[–1.02, +20.01]	[+1.53, +19.69]
11	priorcoach(anneal) – coach(anneal) (held-out, EARLY)	elbow	–10.90	[–22.43, +0.64]	[–20.48, –0.91]
12	[coach(anneal) – blank] @ENDPOINT – @EARLY	elbow	+38.61	[–6.87, +84.08]	[+2.60, +79.94]
13	prior – blank (held-out, MID)	elbow	–12.25	[–25.34, +0.84]	[–22.91, –0.63]
14	coach(anneal) – prior (held-out, LATE)	elbow	+21.52	[–1.06, +44.10]	[+5.97, +46.17]

Table S1. The 14 contrasts of the 134-contrast register on which the primary paired t interval and the BCa bootstrap disagree, and in every one BCa excludes zero and t does not. Under the policy of §3.6 none of them carries a claim anywhere in the paper.

B. Multiplicity: the six family definitions

The uncorrected paired t interval is the primary reported statistic (§3.6). We also computed Holm–Bonferroni at $\alpha = 0.05$, and whether the finger endpoint reversal of §4.5 survives depends on how the contrast family is defined. Over the 28 declared claim-bearing contrasts, 16 survive and the reversal is rank 7. Those 28 are,

however, a 28-cell subset of the 38 t-significant cells §3.6's enumeration rule generates, and correcting inside a set selected on significance understates the correction. Enumerated literally, one contrast per (condition pair × window) among the directly compared conditions, both bodies, plus the two withdrawal-cost differences and the two decay interactions, the rule yields $K = 72$, of which 38 are t-significant, 12 survive, and the step-down stops on the reversal itself ($p = 0.00090$ against a 0.00083 threshold), taking the finger decay interaction, both coach withdrawal costs and every §4.3 endpoint accuracy cell with it. However, that literal set violates §3.6's own exclusion criterion, which forbids cells that sub-divide a window already in the family: $\text{ENDPOINT} \subset \text{LATE}$ and MID overlaps both, and the literal 72 admits all of them. Pruning the duplicated LATE cells gives $K = 58$ (reversal rank 12, survives) and pruning MID as well gives $K = 44$, what the criterion actually derives, on which the reversal is rank 9 against a 0.00139 threshold and survives with room, as do both finger withdrawal cells and the finger decay interaction. A by-question split into route-comparison ($K=26$), versus-model-free ($K=36$) and removal ($K=10$) families also keeps it (rank 5 of 26), and rescues both coach withdrawal costs and two of the four §4.3 endpoint accuracy cells. So the reversal survives four of the six family definitions we report, including the one our own rule derives, and fails the unpruned literal set, which is itself not selection-free. A sixth family we computed cuts the other way; pruning ENDPOINT instead of LATE for the same seven window-spanning pairs also gives $K = 58$, but it removes the title's own cell from the family, so a reader who prefers that prune direction gets no reversal cell to correct and no surviving one (RECOMPUTED.md F1d). Which member of an $\text{ENDPOINT} \subset \text{LATE}$ pair is redundant is a choice, and it determines whether the title claim has a cell at all. We report both directions. What holds under every definition in which it appears, and at every widening of the enumeration we checked, is seven cells: both trained prior – randprior cells, both coach(const) – randcoach cells, the two *finger* coach – blank EARLY cells, and the elbow coach – prior EARLY lead. Four of the seven are adversarial controls and two are the sanity checks of §5. One is a route comparison. The literal enumeration and the summary table of Table S2 follow, carrying one row for each of the five families that contain the reversal's cell, with each family's K , its t-significant count, its Holm survivor count and its verdict on the reversal.

The literal enumeration of §3.6's rule is: seven directly-compared condition pairs × four windows × two bodies (56 cells: prior – blank held-out and train, coach(const) – blank, coach(anneal) – blank, coach(const) – prior, priorcoach(const) – prior, priorcoach(const) – coach(const)); two randprior pairs at $\text{ENDPOINT} \times$ two bodies (4); four withdrawal pairs at $\text{ENDPOINT} \times$ two bodies (8); the withdrawal-cost difference (2); and the decay interaction (2), $K = 72$. That enumeration predates the randcoach control (§3.2), and entering its two pairs on the same footing as the other directly-compared pairs adds $2 \times 4 \times 2 = 16$ cells: $K = 88$ literal, and 52 under the $\text{MID} + \text{LATE}$ prune §3.6's criterion derives. Both verdicts are unchanged, the reversal still fails the literal set, and still survives the derived one at rank 12 against $0.05/41 = 0.00122$, and both coach(const) – randcoach EARLY cells ($t = 7.81$ and 7.15 , $df = 11$) enter above the reversal in every family we computed, so they survive wherever anything does. The table below is printed at the pre-randcoach K . The recomputed values are the ones in this paragraph. Five of the six definitions we report are tabulated below with their verdict on coach(const) – prior ENDPOINT finger ($p = 0.00090$). The sixth, the mirror prune that removes that cell from the family altogether, is described in §4.11 and RECOMPUTED.md F1d and has no row here because the cell it would be scored in does not exist:

family	K	t-significant	Holm survivors	verdict on the reversal
declared (the claim-bearing set: the t-significant subset, minus ten)	28	28	16	survives — rank 7, threshold 0.00227
literal (the enumeration, unpruned)	72	38	12	fails — the stopping cell, threshold 0.00083
LATE pruned (partial prune of $\text{ENDPOINT} \subset \text{LATE}$)	58	31	12	survives — rank 12, threshold 0.00106
MID + LATE pruned (what §3.6's criterion derives)	44	25	12	survives — rank 9, threshold 0.00139
by-question split, route-comparison family	26	19	8	survives — rank 5, threshold 0.00227

Table S2. Five of the six family definitions, with each family's K , its count of t-significant cells, its Holm survivor count and its verdict on coach(const) – prior at ENDPOINT on finger ($p = 0.00090$). The table is

printed at the pre-randcoach K; the recomputed values are in the preceding paragraph. The sixth definition, the mirror prune, has no row because the cell it would be scored in does not exist.

The other two by-question families are versus-model-free ($K = 36$, 12 survive) and removal ($K = 10$, 3 survive); $26 + 36 + 10 = 72$, and 23 cells survive across the three against 12 under the single literal family. Holm is valid under arbitrary dependence; consequently, including correlated cells is conservative and not erroneous, and the literal-72 verdict is correct *about that family*. It is not, however, a selection-free enumeration, because it admits coach abrupt – blank for both bodies while admitting prior strict-withdrawn – blank for neither, dropping the non-significant member of a matched pair. Widening it matters for the survivor count but not for the stopping cell, as rank 12 of the literal family clears its threshold by only $1.12\times$, so eight further less-significant cells take the both-family core from six to five. Cell-by-cell tables for all five families, the K-sensitivity sweep and the three by-question families are in RECOMPUTED.md Appendices C–G.

C. myoHand

Table S3 gives prior – blank on train targets, paired over the 6 shared seeds, positive = the prior is worse. $N = 6$, so the multiplier is $t(.975, 5) = 2.5706$.

window	blank (mm)	prior (mm)	prior – blank	paired t 95% CI	t p	exact Wilcoxon p	seeds on the sign
EARLY {2,4,6k}	106.29	105.63	–0.66	[–7.40, +6.08]	0.812	1.000	3/6
MID {6,8,10k}	108.21	110.08	+1.87	[–6.81, +10.55]	0.604	0.688	3/6
LATE {10,12k}	105.30	115.71	+10.42	[–5.39, +26.22]	0.151	0.219	4/6
12k	106.12	118.61	+12.49	[+0.99, +23.99]	0.038	0.03125	6/6
60k (this run's endpoint)	106.34	99.55	–6.79	[–22.45, +8.86]	0.315	0.563	4/6

Table S3. myoHand, prior – blank on train targets, paired over the 6 shared seeds, positive meaning the prior is worse, with $N = 6$ so the multiplier is $t(.975, 5) = 2.5706$.

Levels at 2k → 12k → 60k: blank 102.5 → 106.1 → 106.3; prior 103.1 → 118.6 → 99.6. Within arm across the budget, blank 60k – 2k = +3.88 [–7.10, +14.86], prior –3.51 [–13.81, +6.80]. On mean(last 5 checkpoints) – mean(first 5), blank –0.28 [–6.28, +5.71] and prior –3.82 [–12.48, +4.83].

Holm over these five window cells, the unpruned window family for this body; the two analysed bodies are corrected under the MID+LATE-pruned $K = 44$ family of §4.11: rank 1 is 12k at $p = 0.03832$ against $0.05/5 = 0.01000$, it fails, and the step-down stops there, so no cell survives. The per-checkpoint grid, computed as well and reported for disclosure and not used, has two of the thirty archived checkpoints reaching $p < 0.05$ (12k at 0.0383, 38k at 0.0359) against a chance expectation of 1.5, and Holm over that grid also stops at rank 1.

D. Additional contrasts

Teacher decontamination (clean – contaminated; positive = decontamination made the student worse). On elbow all eight cells are non-significant, including held-out ENDPOINT (+2.34 [–17.10, +21.77]) and EARLY (+1.85 [–5.20, +8.90]). On finger decontamination significantly deflated the early advantage in all four EARLY cells (coach +15.21 [+9.43, +20.98] held-out and +23.76 [+16.93, +30.59] train; priorcoach +13.72 and +27.29) while leaving the held-out endpoint non-significant (+18.48 [–21.11, +58.07]). The finger anti-synergy survives decontamination, since priorcoach(anneal) – prior at ENDPOINT is +23.99 [+5.46, +42.52] held-out and +26.31 [+9.46, +43.17] train.

Coach schedule. coach(const) – coach(anneal), held-out: ENDPOINT elbow –21.94 [–40.88, –3.01], finger –12.38 [–39.89, +15.13]; EARLY elbow –10.76 [–15.09, –6.42], finger –0.21 [–4.28, +3.86]; MID elbow –36.81 [–51.81, –21.82], finger –14.87 [–27.51, –2.23]; LATE elbow –33.35 [–54.23, –12.47], finger –15.42 [–41.16, +10.32].

Combination, full window table. priorcoach(const) – prior: EARLY elbow –59.30 [–80.25, –38.34], finger –16.54 [–20.62, –12.46]; MID elbow –24.58 [–33.48, –15.68], finger –9.51 [–16.24, –2.78]; LATE elbow –11.43 [–18.43, –4.44], finger +16.23 [+3.83, +28.64]; ENDPOINT elbow –11.53 [–17.80, –5.27], finger +25.63 [+8.88, +42.38]. priorcoach(const) – coach(const): EARLY elbow –8.74 [–17.09, –0.38], finger –4.82

[−10.18, +0.54]; MID elbow −2.12, finger −0.42; LATE elbow +0.40, finger −3.72; ENDPOINT elbow −0.64 [−4.05, +2.77], finger −6.64 [−14.71, +1.43].

Mixture midpoint. priorcoach − 0.5·[prior + coach], where a deviation from zero rejects a 50/50 mixture account: constant schedule, ENDPOINT elbow −6.09 [−8.90, −3.27], finger +9.50 [−1.02, +20.01]. EARLY elbow −34.02 [−45.67, −22.36], finger −10.68 [−13.57, −7.79]. Annealed schedule, EARLY elbow −30.80 [−43.39, −18.21], finger −11.17 [−14.58, −7.76]. Both annealed ENDPOINT cells non-significant.

Tapered-schedule route contrast. coach(anneal) − prior, held-out: EARLY elbow −39.80 [−61.37, −18.24], finger −11.51 [−18.31, −4.70]; MID elbow +14.35 [−0.83, +29.53], finger +5.78; LATE elbow +21.52 [−1.06, +44.10], finger +35.38 [+6.76, +64.00]; ENDPOINT elbow +11.05 [−8.89, +30.99], finger +44.65 [+12.12, +77.18]. Per §3.2 these cells measure a coach whose weight has been falling since step 0 and support no claim.

prior − blank in the intermediate windows. MID elbow −12.25 [−25.34, +0.84], finger −1.32. LATE elbow −21.92 [−32.85, −10.98], finger −12.56 [−37.94, +12.81]. ENDPOINT $\subset$ LATE, so the elbow LATE cell is not independent confirmation of §4.3; coach−blank window interactions (positive = the gap in the aid's favour is smaller later): [coach(const) − blank]@LATE−@EARLY elbow +25.28 [−9.34, +59.90], finger +24.99 [+6.33, +43.65]; [coach(anneal) − blank]@LATE−@EARLY elbow +47.88 [+3.26, +92.50], finger +40.20 [+11.06, +69.34].

E. Additional figures

The figure set is in `wm_prior/figs/`. Every contrast error bar is the paired t 95% CI ($df = 11$), except Figure S1's dispersion bars, which are percentile bootstrap intervals (§4.8); per-condition bars and learning-curve bands carry the corresponding one-sample t 95% CI; the BCa bootstrap is used only to assign flags. Elbow panels are mrad and finger panels are mm, no panel compares them numerically, and `make_figs.py` verifies every plotted contrast against the recomputed table, value, interval and flag, before any image is written, except the four train-target coach(anneal) − blank cells, which are not in that table and are computed with the same estimators and flagged as such. Beyond the five figures in the body: `fig1_2x2_endpoint.png` endpoint bars, train and held-out, both schedules; `fig2_paired_contrast.png` coach − prior at ENDPOINT and EARLY under both schedules; `fig3_guidance_coachVblank.png` coach − blank across target sets and schedules; `fig4_heldout_vs_train.png` prior − blank on both target sets; `fig6_timecourse.png` coach − prior by window under both schedules; `fig8_plateau.png` the $n=3$ long-budget runs, individual seeds shown with no interval drawn; `fig9_mixture.png` the annealed-schedule mixture test; `fig11_coach_dominance.png` the combination against the prior alone, the coach alone and the 50/50 midpoint under the constant schedule. These appendix figures carry the per-cell Holm annotations that the body figures omit.

F. myoArm, the fourth body

`gonogo_myioarm.py` calls `gym.make("myoArmReachRandom-v0")` and trains this study's SAC on it (myoArm: obs 80 act 34 muscles 34). `gonogo_myioarm.log` records greedy fingertip-target distances falling from an initial ≈ 0.98 m to 0.417, 0.352, 0.564, 0.313 and 0.281 m at 2k–10k steps, at which point the run was killed against a declared budget of 40k. It is a single seed, blank only, at hidden=256 in place of the study's 128, under MyoSuite's native reward rather than this paper's reach reward, with no held-out target set, so it supports no contrast, no interval and no claim, exactly as this paper says of myoHand. It is reported because the body count belongs in the paper and because the run writes stdout only and creates no results directory, which is why the completeness register that enumerates results* directories could not have found it (RECOMPUTED.md H4, H5).

G. Seed-to-seed dispersion, in full

A separate, declared apparatus governs the post-hoc dispersion analysis of §4.8. A dispersion is not a paired difference, so the paired t interval above cannot be applied to it and the two estimators above are not what §4.8 uses. §4.8 reports per-condition endpoint SD and CV together with the ratios SD(blank)/SD(condition) and CV(blank)/CV(condition), each with a percentile bootstrap 95% CI (20 000 resamples). Where it compares two dispersions instead of listing them side by side, it uses two further tests, declared here. The first is a Pitman–Morgan paired variance-ratio test over the seeds the two conditions share. The second is a paired bootstrap of the ratio-of-ratios (20 000 resamples), whose interval is read against 1. "Significantly tighter" in §4.8 means exactly this, a ratio-of-ratios interval excluding 1 with an agreeing Pitman–Morgan p , and nothing in §4.8

licenses a claim under any other estimator. These cells sit outside every multiplicity family in this paper for the same reason the ratios do; they are not paired-difference contrasts and carry no paired-t p-value the Holm procedure could take, and their exclusion is not presented as evidence that they would survive one. Everything in §4.8 is post hoc.

The subsection this supports is §4.8. Everything it summarises is here.

This analysis was not planned. It was prompted by a question about what the withdrawal nulls mean, it tests no part of the thesis, and it is reported as a secondary observation and is not a result the paper argues from. We report per-condition held-out endpoint SD across the 12 seeds and the ratio $SD(\text{blank})/SD(\text{condition})$, with percentile bootstrap 95% CIs (20 000 resamples). A ratio above 1 means the condition is *less* dispersed across seeds than model-free SAC. Because reach error is bounded below by zero, SD tracks the mean, so the scale-free CV ratio is reported in Table S4 alongside every SD ratio.

condition	body	mean	SD	CV	SD ratio	95% CI	CV ratio	95% CI
blank (reference)	elbow	70.73	22.78	0.322	1.00	—	1.00	—
prior kept	elbow	50.02	8.18	0.163	2.78	[1.48, 4.63]	1.97	[1.07, 3.28]
prior soft-withdrawn	elbow	52.11	7.34	0.141	3.10	[1.61, 5.46]	2.28	[1.27, 3.80]
prior strict-withdrawn	elbow	57.19	26.05	0.456	0.87	[0.45, 1.58]	0.71	[0.37, 1.28]
coach constant	elbow	39.13	7.77	0.199	2.93	[1.44, 7.25]	1.62	[0.87, 3.73]
coach annealed	elbow	61.07	30.66	0.502	0.74	[0.37, 1.47]	0.64	[0.35, 1.12]
coach abrupt-withdrawn	elbow	55.13	15.06	0.273	1.51	[0.78, 2.78]	1.18	[0.63, 2.12]
prior + constant coach	elbow	38.49	4.77	0.124	4.77	[2.44, 8.96]	2.60	[1.41, 4.68]
prior + annealed coach	elbow	57.72	13.98	0.242	1.63	[0.85, 2.92]	1.33	[0.72, 2.31]
blank (reference)	finger	117.60	43.03	0.366	1.00	—	1.00	—
prior kept	finger	90.50	25.57	0.283	1.68	[0.54, 3.51]	1.29	[0.42, 2.59]
prior soft-withdrawn	finger	88.12	22.14	0.251	1.94	[0.63, 3.74]	1.46	[0.49, 2.65]
prior strict-withdrawn	finger	104.69	31.98	0.305	1.35	[0.44, 2.68]	1.20	[0.40, 2.40]
coach constant	finger	122.77	12.18	0.099	3.53	[1.16, 6.67]	3.69	[1.28, 6.60]
coach annealed	finger	135.15	46.10	0.341	0.93	[0.30, 2.83]	1.07	[0.38, 2.64]
coach abrupt-withdrawn	finger	169.14	34.52	0.204	1.25	[0.41, 2.69]	1.79	[0.64, 3.39]
prior + constant coach	finger	116.13	9.19	0.079	4.68	[1.53, 8.78]	4.63	[1.62, 8.04]
prior + annealed coach	finger	114.49	17.84	0.156	2.41	[0.78, 4.80]	2.35	[0.80, 4.52]

Table S4. Per-condition held-out endpoint SD and CV across the 12 seeds, with the ratios $SD(\text{blank})/SD(\text{condition})$ and $CV(\text{blank})/CV(\text{condition})$ and their percentile bootstrap 95% CIs (20 000 resamples). A ratio above 1 means the condition is less dispersed across seeds than model-free SAC.

Table S5 gives the train-target block, scored on the 8 training targets against the matched train-target blank.

condition	body	mean	SD	CV	SD ratio	95% CI	CV ratio	95% CI
blank (train reference)	elbow	65.65	20.95	0.319	1.00	—	1.00	—
prior kept (train)	elbow	40.15	6.27	0.156	3.34	[2.12, 5.45]	2.04	[1.26, 3.54]

randprior (train)	elbow	89.40	25.08	0.281	0.84	[0.53, 1.27]	1.14	[0.70, 1.87]
blank (train reference)	finger	123.25	38.72	0.314	1.00	—	1.00	—
prior kept (train)	finger	88.97	25.19	0.283	1.54	[0.78, 2.84]	1.11	[0.55, 2.09]
randprior (train)	finger	129.78	11.65	0.090	3.33	[1.66, 7.59]	3.50	[1.81, 7.66]

Table S5. The same quantities scored on the 8 training targets, against the matched train-target blank, and held-out and train ratios are never mixed.

An aid still attached at 12k compresses seed-to-seed dispersion in three of the four kept-aid cells on the SD ratio and in two of the four on the scale-free CV ratio. On SD: elbow prior 2.78, elbow coach 2.93 and finger coach 3.53 exclude 1, and the fourth cell, finger prior kept at 1.68 [0.54, 3.51], does not. On CV: elbow prior 1.97 and finger coach 3.69 exclude 1, and elbow coach falls to 1.62 [0.87, 3.73], joining the finger prior's 1.29. The two combined priorcoach(const) arms carry the study's numerically largest ratios (4.77 [2.44, 8.96] elbow, 4.68 [1.53, 8.78] finger), though, tested on the axis declared at the head of this appendix, neither is distinguishable from the next cell on its body (elbow against prior soft-withdrawn, ratio-of-ratios 1.54 [0.83, 2.80], Pitman–Morgan $p = 0.190$; finger against coach constant, 1.33 [0.87, 2.04], $p = 0.362$), so the ranking is reported, not claimed. After the two step-function withdrawals, the strict prior and the abrupt coach, dispersion is not distinguishable from the model-free baseline (elbow strict prior 0.87, elbow abrupt coach 1.51), a result that is underpowered in both directions. This does not generalize to withdrawal as such, and the elbow soft prior withdrawal, which differs from the strict one only in that the imagined-transition buffer is not purged, is at 3.10 [1.61, 5.46] the tightest of the three withdrawn elbow arms. Two SD point estimates are not a comparison, so that row is tested on the same axis as the rest of this subsection, over the twelve shared held-out seeds, where the soft arm is significantly tighter than both step-function withdrawals, against the strict prior withdrawal, ratio-of-ratios 3.55 [2.27, 6.11], Pitman–Morgan $p = 0.00042$; against the abrupt coach withdrawal, 2.05 [1.19, 3.68], $p = 0.03204$, and it is not distinguishable from the kept arm it was withdrawn from: 3.10 against the kept arm's 2.78 is a ratio-of-ratios of 1.114 [0.680, 1.896], an interval including 1, with Pitman–Morgan $p = 0.70198$, and the finger twin is 1.155 [0.809, 1.546], $p = 0.51932$. Read as two point estimates, the 3.10-versus-2.78 gap looks like an inversion of the expected direction. Tested on the axis this appendix declares for dispersion comparisons, it is not distinguishable from 1 (1.114 [0.680, 1.896], $p = 0.70$): the inversion is not established, and at this power neither is its absence.

The dispersion effect is not attributable to what the model knows, on finger. The adversarial control answers the objection's twin, that any frozen model in a Dyna loop might contract spread regardless of content. On elbow the uninformative model does not suppress dispersion (0.84 [0.53, 1.27]) while the trained prior on the same targets is at 3.34. On finger the ordering inverts, and randprior is at 3.33 [1.66, 7.59] against the trained prior's 1.54 [0.78, 2.84] on the identical target set, in an arm with no teacher at all. Because two separate interval verdicts do not constitute a comparison, the difference is tested directly over the twelve shared train-target seeds, where a Pitman–Morgan paired variance-ratio test gives finger $p = 0.02117$, and a paired bootstrap of the ratio-of-ratios gives 2.163 [1.067, 5.130], excluding 1. The elbow twin is significant in the opposite direction (0.250 [0.149, 0.386], $p = 0.00010$). So the two bodies' point verdicts run in opposite directions on content-specificity. Bodies are never pooled inside a contrast (§3.6), so no test across bodies is available and the disagreement is described but not established. On myoFinger the content-specificity ordering runs opposite to elbow's, and we do not have an account of it. Both cells are post hoc and are ratios of sample variances rather than paired-difference contrasts. Two further mundane rivals are reported rather than argued away, as the CV ratios are smaller than the SD ratios, so the effect is partly a consequence of the mean differences reported separately. Moreover, a shared-artefact rival applies to both routes equally, with all twelve coach seeds distilled toward one frozen teacher and all twelve prior seeds handed one frozen world-model checkpoint, so contraction is partly mechanical in both.

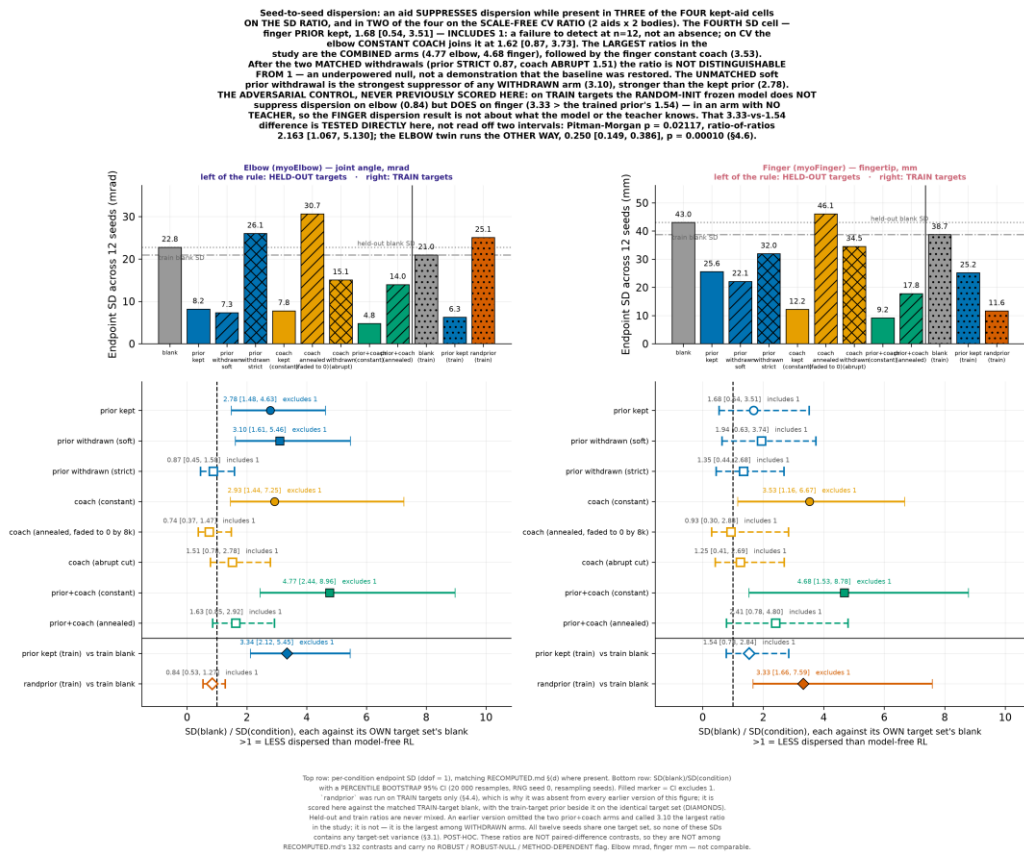


Figure S1. Per-condition endpoint SD across 12 seeds and the SD(blank)/SD(condition) ratios with percentile bootstrap 95% CIs, for the nine held-out conditions tabulated in Table S4 per body (randcoach, added late, is in §4.1 but not that table), plus the three train-target rows of Table S5 scored against the matched train-target blank, and held-out and train ratios are never mixed (fig13_variance.png).

H. The remaining limitations

These are the eight §6 defers. Each is stated in full; none of them is load-bearing for a claim in §4 or §5, which is why §6 carries the other seven instead.

Claims withdrawn during the study. Five claims were withdrawn over the course of the work, in every case after the control or the direct paired contrast that tested them was run: a double dissociation between the two routes, a back-loading of the prior's advantage, a withdrawal asymmetry between routes, a decay of the prior's contribution to the combination, and a dispersion "inversion" that is a null (§4.8). Separately, an interval-estimator error, a normal-approximation multiplier ($z = 1.96$) where a t multiplier is required at $n=12$, was corrected by recomputing all 132 contrasts under the policy of §3.6, a correction that could only remove significance and never create it. The itemised record, every withdrawal, every recomputation, the scope and body-count corrections behind §4.10 and Appendix F, and the unarchived build transcripts of §3.3, is in RECOMPUTED.md, and any completeness statement about this repository is scoped there to *no unreported artefact remains that writes a file into this tree*, with named exclusions. Thirty-two further 12k-step runs sit under actors_withdraw/ and actors_randcoach/ (twelve, at two seeds), actors_finger/ (sixteen, four arms x four seeds on myoFinger, logs in logs_fingervid/) and actors_video/ (four, myoElbow seed 0), all trained after this analysis was complete solely to render the comparison videos. Twenty-seven re-execute conditions already reported, bit-identically where we have checked (video-run finger blank at 12k is 78.4 / 107.8 / 142.7 / 106.8 against the study's 78.4247 / 107.8350 / 142.7321 / 106.8157). The other five are the study's only held-out scoring of randprior, one on myoElbow at seed 0 (actors_video/randprior_actor.pt, evaluation output not redirected to a file and unrecoverable) and four on myoFinger at four seeds, giving randprior – blank = +37.30 [–31.00, +105.60] and randprior – prior = +63.70 [+11.26, +116.14]. They were not planned as an arm, the finger four are $n=4$, and they carry no claim.

- The obvious rival reading of the myoHand null is one these data cannot settle. A body model might be expected to pay off *more* on a harder body, not less, and myoHand is the test: 23 joints and 39 muscles against

myoFinger's 4 and 5. However, neither arm improves there within 60k steps, so there is no advantage to amplify and nothing to measure. Whether the larger bodies fail because the method does not scale, or because the budget does, is a question this study cannot separate. The literature it opens against trains for hundreds of millions of steps and the hardest body here received 6×10^4 . The hypothesis is untested at this budget and is not refuted by it.

- The frozen world model is a single checkpoint per body, loaded frozen and handed identically to all twelve seeds, so every prior, randprior and priorcoach interval is conditional on one artefact whose seed variability is unquantified.
- Seven analysed conditions (fourteen directories) have no archived launcher (`results_noanneal_*`, `results_abruptcoach_*`, `results_cleancoach_*`, `results_cleancoach_train_*`, `results_pconst_*`, `results_prioroff_*`, `results_priorpurge_*`). Their invocations are reconstructible from the run log and checkpoint structure but not from a committed script. Two are provable from the data itself on myoElbow, where the constant and abrupt coach arms agree to ≤ 0.02 mrad before the cut and a teacher swap costs ≈ 5.8 mrad at 2k (on myoFinger the same pair differs by up to 1.67 mm pre-cut, small beside the +46.37 mm withdrawal cost but not a bit-identity), but the constant coach's teacher identity relative to the annealed coach rests on the run log alone, and §3.2's `coach(const) - coach(anneal)` contrast depends on it.
- There is one teacher per body (seed 100), trained with 30,000 rewarded task-specific steps that no other arm receives and saved as a terminal checkpoint off a non-monotone curve with no model selection, so the teacher-ceiling account of §4.5 is untested and every teacher-dependent claim is scoped to the teacher as captured.
- All twelve seeds share one training target set and one held-out target set, so held-out generalization rests on a single draw of 20 targets and the dispersion analysed in §4.8 contains no target-set variance at all.
- No hyperparameter search was run, so baseline-tuning parity between blank and the aided arms is assumed but not established, and the omission is not neutral between the routes, since the coach has one free parameter against the prior's five untuned Dyna knobs.
- No behavioural variable beyond distance and return was logged, no activation magnitude, path length, terminal velocity or per-target error distribution, which is the cheapest missing diagnostic in the study.
- The guidance route's adversarial control was added late (§3.2, §4.2): randcoach runs from step 0, so it excludes "any fixed, non-flailing action target lowers early error" for the kept aid. Furthermore, it was not run as a *swap at 8k*, so the withdrawal cells of §4.4 still have no matched random-teacher comparison.

I. Protocol and implementation details

The material §3.1, §3.3, §3.4 and §3.5 defer, verbatim.

Evaluation protocol. Actions are greedy (deterministic); the start pose is fixed; each episode is 100 environment steps, one episode per target; there are 8 training targets and 20 disjoint held-out targets; the reported distance is the final-step distance averaged over targets, and teachers, students and referents are scored identically. `evaluate()` in `run_reach.py` logs exactly two quantities per checkpoint, and both are used in this paper. `eval_dist` is that reported error, the final-step target distance averaged over targets, in mrad on myoElbow and mm on myoFinger. `eval_return` is the undiscounted sum of the environment's per-step task reward over the 100 steps of the same greedy episode, averaged over the same targets. On myoFinger the per-step reward is $r = -\text{dist} + 0.1 \cdot 1\{\text{dist} < 10 \text{ mm}\} - 0.001 \cdot \|\text{ctrl}\|^2$ (`myofinger_env.py`), so `eval_return` is in the same length unit as `eval_dist` multiplied by steps, and $-\text{eval_return}/100$ is a mean within-episode error in `eval_dist`'s own units, up to the small target bonus and control-effort terms. §5's movement index is exactly the difference between that mean and the terminal error. The target sets are fixed global constants shared by all twelve seeds. The run seed enters only through network and environment initialization and the order in which training targets are sampled at reset, so held-out generalization rests on a single draw of 20 targets and the seed-to-seed dispersion of §4.8 contains no target-set variance.

Backbone. One SAC backbone, hidden width 128, update-to-data ratio 2, LayerNorm critics, gradient clipping (norm 10), entropy target $-0.5 \cdot |A|$, lr $3e-4$. Dyna (used by prior, randprior, priorcoach): rollout horizon $k = 3$, `dyna_every = 100`, `dyna_starts = 200`, batch 128, `real_ratio = 0.5`. This is the single set of defaults in `run_reach.py`, fixed once at the start of the study and applied unchanged to every condition, seed and body that carries analysis. The myoArm probe of Appendix F used hidden=256 and MyoSuite's native reward instead, and the colearn arm of §4.7 deviates in the ways enumerated there, a different learning rate, re-initialisation at every refresh, refitted normalizers, a validation holdout and a periodic RNG reset. No hyperparameter search was run at any point.

The prior is an MLP $(s,a) \rightarrow \Delta s$, trained on task-agnostic random (Ornstein–Uhlenbeck) muscle activations with no task and no reward, then reused frozen. Open-loop accuracy: myoElbow $\approx 1.1^\circ / 0.019 \text{ rad}$ ($\approx 19 \text{ mrad}$) joint error; myoFinger 14.3 mm mean, 9.8 mm median 5-step fingertip error; myoHand 8.5 mm mean, 7.5 mm median (on 3×10^5 babble transitions). The archived build transcripts give 2×10^5 transitions for myoFinger (39 s of collection on CPU) and 3×10^5 for myoHand (31 min), so the two archived counts span 2×10^5 – 3×10^5 and neither speaks to the third body, myoElbow. The elbow prior has no archived build transcript at all, and prior_my0.pt is on disk and is loaded by every elbow prior, randprior and priorcoach run in the study, but no log of its babbling or of its open-loop evaluation exists anywhere in the repository, so both the $\approx 19 \text{ mrad}$ fidelity figure and the elbow babble budget are unarchived point values, and the fidelity figure is load-bearing in §5's third qualification of the forward-model arm. All three fidelity figures are measured on babble-distribution data and are never re-measured on the on-policy state distribution, so they are an optimistic bound on fidelity in the regime where the model is queried. One checkpoint per body is loaded frozen and handed identically to all twelve seeds.

Teachers are trained from blank (model-free SAC, no body prior anywhere) for 30,000 rewarded, task-specific environment steps, $2.5 \times$ a student's entire budget, at a single seed (100) per body. train_teacher2.py writes exactly one checkpoint, at the end of training, with no model selection. Held-out teacher scores: elbow 27.2 mrad, far better than any student; finger 169.1 mm, worse than every student condition mean bar one, the withdrawn finger coach (169.14) being numerically identical to it and just under the no-op referent of 172.3 mm. Both saves come off violently non-monotone curves, and the elbow save lands on the global minimum of its own train-target curve (which spans 15.5–149.0), while the finger save (133.5 on train targets) is worse than each of the three checkpoints before it, on a curve spanning 102.9–207.9. The 12k checkpoints of the same two runs, scored with the unmodified teacher_eval.py, reach 62.6 mrad and 129.4 mm held-out. Every claim resting on teacher quality is therefore scoped to *the teacher as captured*, namely the seed-100 checkpoint at exactly 30,000 steps. Teachers produced by invoking the prior condition carry the body prior in laundered form. All coach and prior+coach cells reported here use clean teachers, with one labelled exception: Appendix K quotes the contaminated 2×2 coach's train-target endpoint (+9.10) beside the clean annealed cell, because the constant schedule was never run on train targets at all. The decontamination contrasts are in Appendix D.

Neither intervention arm is matched to blank, and the two routes are not matched to each other. The prior arm consumes task-agnostic, reward-free babble transitions of the least task-relevant kind before its 12k-step run, and the archived counts are the ones quoted in §3.3: 2×10^5 transitions on myoFinger and 3×10^5 on myoHand, with no archived transcript at all for myoElbow. On myoFinger, therefore, the prior arm's babble budget is $6.7 \times$ the 30,000 steps the coach's teacher consumes ($2 \times 10^5 \div 3 \times 10^4 = 6.67$), and that is the one multiple in this paper computed from two archived figures. On myoElbow no multiple can be stated; the numerator is unarchived, so the elbow prior's raw-interaction advantage over the teacher is asserted to be of the same order and is not measured. Wherever this paper says the prior arm is ahead on raw interaction, the supporting arithmetic is the finger one. The coach arm's teacher consumes 30,000 rewarded steps *on this very reach task*, which is directly about the scored quantity. The prior arm is thus simultaneously ahead on raw interaction, behind on task-relevant interaction, and behind on real-data gradient signal by $2 \times$ per update (§3.1). No experiment here equalizes any of the three.

I.1 Withdrawal mechanics and the statistical rule

The two paragraphs §3.4 and §3.6 defer, verbatim.

Each aid is cut at env-step 8000 and every condition is scored at 12000. For the coach this requires the abrupt schedule. For the prior we distinguish soft withdrawal (stop generating imagined transitions but continue sampling those already buffered) from strict withdrawal (stop generating *and* purge the buffer). Strict was intended as the matched control, for a design reason, because the coach condition has no imagined-transition buffer at all, so for the prior arm to be matched, removal must also leave the agent training on real data only. As implemented it is not matched, and the mismatch is in exactly that respect. Purging sets the imagined buffer's count to zero, and rollouts also stop at the cut, so the buffer never refills. The update rule's guard (`model_buf.n < batch`) is then permanently satisfied and the strict arm silently switches from 64 real + 64 imagined to 128 real per update for the rest of training. The prior's strict withdrawal therefore changes two things at 8k, imagination is removed *and* the real-data gradient signal doubles, while the coach's abrupt withdrawal changes one. The soft arm keeps the mix at 64 + 64 but on a frozen, stale imagined buffer, so it is unmatched in the other direction. Neither withdrawal is a clean single-factor manipulation, and we report both with that stated.

The primary statistic is the paired t 95% CI, $\text{mean}(d) \pm t_{0.975, n-1} \cdot \text{sd}(d)/\sqrt{n}$, reported as an effect size with its interval. A contrast is reported as significant only when that interval excludes zero. Two robustness checks accompany it and neither is used to license a claim, a BCa bootstrap 95% CI (20 000 resamples, RNG seed 0, resampling the vector of paired seed differences) and an exact Wilcoxon signed-rank test. Two re-scalings of the decay interaction are declared here, used only in §4.5 and §7 and licensing no separate claim. $[\text{coach}(\text{const}) - \text{prior}]@LATE - @EARLY$ is a difference of differences of a quantity bounded below by zero, measured at two very different error levels, so it is reported under three scalings: absolute, $(c_L - p_L) - (c_E - p_E)$, in mrad or mm; log-error, $(\ln c_L - \ln p_L) - (\ln c_E - \ln p_E)$, a change in the *ratio* between the two arms; and blank-normalised, each arm divided by that seed's own blank in the same window before differencing, so the units are fractions of the untreated baseline. All three use the paired t interval above. Where they disagree, §4.5 reports the disagreement and does not choose among them. Over the 134 contrasts computed, 60 are significant under t , 60 under neither estimator, and 14 under BCa alone. The t -only cell is empty; thus BCa is uniformly the more permissive side. The 14 BCa-only contrasts are tabulated in Appendix A and carry no claim. Per-checkpoint cells sit outside every correction family reported here, because they sub-divide windows already in the family and correcting them would need a scheme we have not run. The per-checkpoint finger cells of §4.5 are therefore asserted uncorrected, and §4.10 records that the exemption was kept even where entering the cells would have helped us. All contrasts are computed on unrounded per-seed data and may differ from differences of displayed rounded means by ~ 0.1 unit. The full recomputation of every contrast outside the randcoach arm is RECOMPUTED.md. Its 134-contrast register carries both estimators. The contrasts added later carry the primary estimator only: the two non-absolute scalings of the decay interaction per body (§4.5), the frozen12 cells (§4.7), every myoHand cell (§4.10), the train-target coach cells (§4.3) and the per-checkpoint cells (§4.5), as do the randcoach contrasts of §4.2, which `analyze_randcoach.py` computes.

J. The remaining next steps

§7 names the first four and, out of order, blank to 30k. This is the full list as it stood at submission.

Next steps. Log at least one behavioural variable, mean activation, path length, terminal velocity, per-target error, at every evaluation checkpoint, and resample the target sets per seed. Then, in rough order of cost, we would run the swap half of the randcoach control, whose from-step-0 arm is reported in §4.2. Substituting a random teacher *at the 8k withdrawal point* would separate the teacher's content from any fixed non-flailing target in the withdrawal cells, and would separate the withdrawal cost from the optimization shock. After that we would add a matched-replay Dyna control that swaps the imagined half of each batch for replayed real transitions, and run warm and colearn at $n=12$ on myoFinger and on held-out targets; the myoElbow train-target half is now run (§4.7), and the remaining halves would test whether the tight null and the babble/dispersion effect replicate. A withdrawal experiment sized against the *direct cost-difference* contrast would need more seeds than we ran. Multi-seed and multi-checkpoint clean teachers, since running a student against the 12k teacher checkpoints already on disk is a teacher-quality manipulation available at zero training cost. Intermediate distillation weights. blank to 30k at $n=12$, which would settle the amortization reading. A matched-total-interaction baseline. And controls breaking the information asymmetry in both its parts: a task-agnostic coach or a task-aware body model, and a Dyna variant with a *learned* reward model in place of the analytic oracle.

K. The forward-model qualifications, the amortization series and the movement index

The full versions of the material §4.2, §4.3, §4.4, §5 and §7 summarise.

The two routes can now be compared directly. While present, the coach is the stronger aid at the myoElbow held-out endpoint and faster than the prior on both bodies. That lead decays on both, and inside a 12k budget the decay crosses zero on myoFinger and not on myoElbow. The route comparison is the contrast most exposed to the design asymmetries of §3.1 and §3.5. The coach arm trains on twice the real data per update, while the prior arm consumes far more raw interaction ($6.7\times$ the teacher's 30,000 steps on myoFinger, unarchived and so unquantified on myoElbow, §3.5) and holds a reward oracle the coach does not. When both aids are present we cannot detect a difference between the combination's endpoint and the coach alone's on either body, and after a matched withdrawal no route-level *durability* statement is available. Every claim this paper makes about an aid beating model-free RL at the endpoint is an elbow claim. On finger the prior-versus-blank cells are not significant under the primary estimator, the constant coach beats neither blank nor the prior at 12k, and the

single finger endpoint claim against model-free RL runs the other way (the withdrawn coach is +51.5 worse). That restriction to the endpoint is load-bearing, because the EARLY window runs the other way, and on finger both coach schedules beat model-free RL early (const -17.60, anneal -17.39).

Four qualifications belong in the same breath. The prior arm evaluates the environment's analytic reward at its imagined states, so the endpoint reversal is scoped to a forward model *plus a known reward function* and is as much an oracle result as a forward-model result (§3.5). Cutting the prior is not shown to be costly on either body and the direct removal-cost difference between the routes is non-significant on both, a null with a named candidate generator (§3.4), so nothing here says that what the model contributes is better retained. The decay holds in raw units and under blank-normalisation on both bodies and, on myoElbow alone, not under the log-error reading (§4.5). Finally, the reversal's status under multiplicity correction is family-dependent, since it survives four of the six definitions we report, including the one §3.6's own exclusion criterion derives, and fails the unpruned literal set, while seven cells survive every definition in which they appear (§4.11).

coach(const) - prior in the EARLY window {2,4,6k}, held-out, is -50.56 on elbow and -11.72 on finger, so the constant-weight coach is ahead of the frozen prior early on both analysed bodies, and that lead is bought by what the teacher contains. The adversarial control for this route, randcoach, a random-initialized teacher of identical architecture in the same loss at the same weight, does not reproduce that gain on myoFinger and is significantly worse than no teacher on myoElbow. The direct within-axis contrast coach(const) - randcoach is -19.03 [-24.40, -13.67] on finger and -108.60 [-142.01, -75.18] on elbow; against model-free RL the same arm is +1.43 [-6.03, +8.89] on finger EARLY and +24.41 [-3.16, +51.99] at the endpoint, both failures to detect. The finger endpoint twin is -19.25 [-32.66, -5.84]. No randcoach contrast appears in RECOMPUTED.md; they are computed by analyze_randcoach.py under the primary estimator only (§3.6). §5 records why this control is the one the guidance claim needed. Against model-free RL in the same window, where the aid is unambiguously present: coach(const) - blank elbow -59.03 [-87.11, -30.95], finger -17.60 [-23.93, -11.27]. The tapered coach reproduces the same EARLY effect on both bodies (elbow -48.27, finger -17.39). The prior's own EARLY cell against model-free RL is not significant under the primary estimator on either body (elbow -8.47 [-31.82, +14.88]; finger -5.88 [-16.15, +4.38], BCa-only, Appendix A, the same class §4.3 flags for its siblings), and it decomposes the elbow route contrast exactly: -59.03 + 8.47 = -50.56. On elbow in the EARLY window, therefore, coach - prior is coach - blank plus a quantity we cannot distinguish from zero. This was not the a-priori favourite, as the frozen model supplies exactly what classically pays off earliest [2,6], and the two EARLY nulls above are where that expectation fails (§1).

The prior's elbow effect is $70.7 \rightarrow 50.0$ mrad held-out, -20.7 mrad $\approx -1.19^\circ$, a ~29% reduction in reach error, significant on held-out and on train targets. The coach's endpoint advantage has no train-target twin, because the constant schedule was never run on train targets. The two schedules that were run there, annealed and the contaminated 2x2 coach, do not beat the prior on train targets (+15.24 and +9.10 [-12.61, +30.80], both n.s.). Their point estimates run opposite to the held-out -10.89, and both train-target coach arms use the default linear taper, so their coach weight is already zero from 8k to the 12k scoring point. They are post-withdrawal numbers and not a still-attached twin. We report that asymmetry rather than leaving the prior's corroboration row to stand alone. The constant-weight coach improves elbow endpoint accuracy by more (-31.6 against model-free SAC), and the direct route-versus-route contrast, the quantity a claim about which aid is stronger actually requires, is -10.89 in the coach's favour. While present, the coach is the stronger aid at the myoElbow held-out endpoint. Nothing is claimed for the prior on finger, where the three prior-versus-blank finger cells of Appendix A are significant under BCa but not under the primary estimator, and the finger MID and LATE cells (Appendix D, -1.32 and -12.56 [-37.94, +12.81]) are significant under neither. Two scope conditions travel with these cells. The constant coach's endpoint is scored while the loss is still pulling the student toward a competent teacher, so it is close to definitional taken alone and what makes the pair informative is the contrast *between schedules*; and the two arms are not scored on equal real-data gradient signal (§3.1).

Cutting the coach is costly on both bodies. On myoElbow, where the teacher as captured is competent, that cost is a within-route acquisition-versus-retention dissociation and is the cleanest result in the withdrawal experiment; §5 reads the finger cost differently, since the finger teacher sits essentially at the no-op referent. On finger, removing the coach leaves the agent worse than never having had one (+51.54 against blank) and worse than the withdrawn prior (+64.45), both produced by the withdrawn finger coach landing essentially at the no-op referent rather than by the withdrawn prior being retentive. Cutting the prior is not shown to be costly on either body, and no route-level durability asymmetry is claimed, since the prior's own withdrawal cost is non-significant on both bodies (prior strict-withdrawn - prior kept: elbow +7.17; finger +14.19), and so is the direct route comparison (coach cost - prior cost: elbow +8.83; finger +32.18). Neither null is treated as evidence of retention,

and the first has a named candidate generator, given that by §3.4 the strict arm doubles its real-data gradient signal at the cut, which should *help* it, biasing this very contrast toward zero, and the finger interval misses zero by 0.95 mm. The derived coach cost – prior cost contrasts inherit the same bias, in the direction of inflating the coach's cost relative to the prior's. The elbow soft-withdrawn prior does beat blank (–18.62), but that manipulation is unmatched and biased in the prior's favour, so it is not evidence that the prior persists. The cost does not identify two things at once; cutting the coach at 8k removes the teacher's information *and* deletes a weight- ≈ 1.0 auxiliary loss term mid-training, and Appendix O reads the finger cost as dominated by that optimization shock. Neither prior withdrawal is single-factor either (§6).

That leaves the controls, and what the guidance control settled. The two routes are not equally well controlled: the forward-model route has an adversarial control, randprior, whereas the guidance route had none for most of this study. randcoach, a randomly-initialized frozen teacher distilled with the same auxiliary loss and weight, was added last, and that gap matters most exactly where the guidance result looks strongest. The clean finger teacher as captured scores 169.1 mm, worse than a blank student and within 3.2 mm of the no-op referent, and yet the finger coach still produces a large EARLY lead. So "distil the student toward a near-no-op action target on the body where inaction is worth 71 mm over thrashing" predicts an early gain of the observed sign *independently of anything the teacher encodes about the task*. Without a control, that account could not be separated from ours.

We have now run the control, and randcoach distils a randomly-initialized frozen teacher of identical architecture through the same auxiliary loss at the same constant weight, matched to coach(const) in body, budget, target sets and seeds (§3.2). On the body that raised the rival account, the direct contrast separates the two. On myoFinger EARLY, randcoach – coach(const) is +19.03 [+13.67, +24.40]. That is the within-axis comparison §1 requires, and does not infer one from two separate verdicts. Accordingly, a randomly-initialized teacher is itself a fixed non-flailing action target, and it does not reproduce the coach's early gain. Against model-free RL the same arm is +1.43 mm [–6.03, +8.89], where coach(const) – blank is –17.60 [–23.93, –11.27]. Per §6 that interval is a failure to detect at $n=12$ and is not evidence that a random teacher does nothing.

On myoElbow an uninformative teacher is significantly *worse* than no teacher at every window (EARLY +49.57 [+23.37, +75.76]; MID +21.54 [+1.53, +41.54]; LATE +20.29 [+0.58, +40.00]; ENDPOINT +22.53 [+2.80, +42.26]). That is the shape randprior shows for the forward-model route on the same body at the endpoint, the only window on which it was scored. Both routes now carry an adversarial control, and on myoElbow each is worse than no aid on the target set it was run on: randprior on train (+23.8 [+1.7, +45.8]), randcoach held-out (+49.6 [+23.4, +75.8]). The two controls are parallel in design and are not matched in scoring. On myoFinger randcoach is not distinguishable from model-free RL at EARLY or at the endpoint, and randprior is not distinguishable from it at the endpoint on train targets, the only window and target set on which it was scored; randcoach is significantly worse than model-free RL at MID (+12.86 [+8.12, +17.60]) and LATE (+26.44 [+11.86, +41.02]), the same shape it shows on myoElbow at every window (§4.2, §4.6).

Four qualifications follow from our own controls. First, the *shape* is unusual: on elbow the benefit is not detected EARLY (–8.47 [–31.82, +14.88], an interval that does not exclude an effect the size of the endpoint one), not detected in MID (–12.25 [–25.34, +0.84]; BCa-only, Appendix A, and so carrying no claim), and present LATE and at the endpoint, the reverse of the classical model-based story [2,6]. That shape is three window verdicts, not a tested growth, and the within-seed interaction §3.6 requires for any claim that an advantage grows, [prior – blank]@LATE – @EARLY, is null on both bodies, elbow –13.45 [–39.34, +12.45], finger –6.68 [–36.49, +23.13], so the pattern is described here without being established. The corresponding coach – prior interactions, which are significant, are the ones §4.5 rests on. One plausible but untested mechanism is that the frozen model becomes useful once the policy's state distribution has expanded. A second, entirely mundane one is that this arm is handicapped by 64 real transitions per update throughout, and a handicap of that kind bites hardest early.

Second, the arm operationalizes a forward model plus a known analytic reward function evaluable at imagined states, so any biological reading must assume the nervous system has access to something like a state-evaluable cost at unvisited states. That scope covers the finger reversal as much as the elbow accuracy result, since an arm that can evaluate the true reward at unvisited states has a standing mechanism to keep improving after model-free arms plateau, and the finger prior falls 107.31 $\rightarrow$ 90.50 over exactly the last 2k that carries most of the +32.27. Read that way the reversal is an oracle result as much as a forward-model result, and nothing here separates the two.

Third, the elbow model's own ≈ 19 mrad open-loop prediction error is essentially the size of the -20.7 mrad endpoint effect attributed to its fidelity. That figure has, as §3.3 records, no archived build transcript behind it, and was measured on the babble distribution rather than on the on-policy distribution where the model is actually queried.

Fourth, both endpoint accuracy results have an amortization reading that the available data do not exclude. Four unaided long-budget elbow runs sit on disk, and at each run's true terminal checkpoint, 32k for the three continued blank seeds and 30k for the teacher run, they end at 31.5, 35.3, 41.8 and 27.2. Three of the four are below the constant-coach student's 39.13, and all four are below the kept prior's 50.02, by a wider margin in the prior's case. The same probe on myoFinger points the same way against the coach and not against the prior, and at 32k the three continued seeds are 118.4 / 87.2 / 104.8, plus the teacher run's 169.1 at 30k, three of the four below the constant coach's 122.77, while only one of the four falls below the kept prior's 90.50. Under that reading, on myoElbow, each aid buys a place on the blank learning curve, and does not reach something a longer unaided run would miss. Each of the four is a single terminal checkpoint of an oscillating curve with no interval, and the settling experiment, blank to 30k at $n=12$, was not run.

One thing does constrain it, since the archived `eval_return` makes a movement index recoverable on finger, $M = (-\text{eval_return}/100) - \text{eval_dist}$, whose sign separates "ends nearer than its episode average" from "ends farther". Scored on §4.9's own referents under the identical protocol, the zero-activation no-op is at $M = -40.12$ held-out (-26.89 train), random action -47.03 and the clean finger teacher -17.69 . All eleven finger arms lie in $[-8.16, +2.80]$, each at least 29.7 mm from the no-op referent measured on its own target set. Per-seed SDs run 2.55–11.28, so the separation is $\approx 7.5\sigma$ at the arm with the smallest separation in millimetres and 2.8σ at the noisiest, randcoach ($M = -8.16$, per-seed SD 11.28, 31.96 mm from the held-out no-op referent), which was added late and is computed here rather than in RECOMPUTED.md D1. No finger arm reproduces the zero-activation policy's movement signature. What M cannot do is measure how *much* an arm moves, or discriminate against a constant-activation policy other than the zero-activation one, which is what the rival actually turns on. The randcoach control above settles the rival directly, though a single logged mean activation would still sharpen the mechanism.

L. Competence referents, the long-budget probe, and myoHand

The two subsections §4.9 and §4.10 summarise, in full.

Table S6 gives the absolute competence referents, scored under the identical protocol and target sets.

referent	elbow train (mrad)	elbow held-out (mrad)	finger train (mm)	finger held-out (mm)
random-action policy	634.9	659.8	220.0	243.4
zero-activation (no-op)	676.2	692.4	155.2	172.3

Table S6. Absolute competence referents, a zero-activation no-op policy and a random-action policy, scored under the identical protocol and target sets.

These are unreplicated point estimates, and `random_baseline.py` runs one rollout per target under one RNG seed. Re-run over eight RNG seeds on finger, the random-action referent is 249.2 ± 10.8 mm held-out and 247.4 ± 14.2 mm train, so the archived 220.0 mm train figure is a favourable draw. The elbow referent was not re-drawn. We use solved in exactly one sense: a body is solved by a condition when that condition's held-out error is an order of magnitude below the random-action referent and far below the no-op referent. Only one of the two bodies gives a resolvable ordering between the referents. On elbow random-action (659.8) is nominally below no-op (692.4), a gap of 32.6 mrad against across-target SDs of 460 and 543 mrad on those two draws, roughly 0.2 standard errors, and the elbow referent was not re-drawn, while re-drawing the finger referent under the identical protocol moved it by 12.5% of the archived draw ($220.0 \rightarrow 247.4 \pm 14.2$, itself 1.9 SD above the archived value), so we report the elbow numbers and read no ordering from them. On finger the gap is far larger and runs the other way, with no-op 172.3 beating random-action 243.4 by 71 mm, so on myoFinger inaction is a good policy and thrashing an actively bad one, and the clean finger teacher as captured (169.1 mm) sits 3.2 mm below the zero-activation referent. Distilling toward it is therefore close to distilling toward "stop moving" on the body where stopping is good (§5).

Eight of the ten elbow conditions are solved in that sense. Model-free blank ($9.33\times$) and the random-teacher control randcoach ($659.8/93.26 = 7.07\times$) are not. Ratios against the 659.8 referent: priorcoach(const) $17.1\times$, coach(const) $16.9\times$, prior kept $13.2\times$, prior soft $12.7\times$, coach abrupt-withdrawn $12.0\times$, prior strict-withdrawn

11.5 $\times$, priorcoach(anneal) 11.4 $\times$, coach annealed 10.8 $\times$, and blank at 70.73 mrad is 9.33 $\times$, which is not an order of magnitude, a miss of 6.7%, well inside the imprecision of a single unreplicated draw. Every elbow contrast in this paper is a difference *within* that regime, on a 1-DoF body whose held-out targets are near-interpolation. myoFinger is solved by no condition; the two lowest-error finger conditions are the soft-withdrawn prior at 88.1 mm and the kept prior at 90.5 mm, only $\approx 1.9\times$ below doing nothing, and model-free SAC (117.6) is only $\sim 32\%$ better than doing nothing. A held-out fingertip error of 90.5 mm is comparable to the finger's own length scale, so the finger endpoint result says which of two poor conditions is less poor.

A supplementary longer-budget experiment (blank and prior, $n=3$ seeds, curves parsed from logs, no intervals) supports no claim. The runs were launched for 36k and killed at 32k, except prior seed 2, whose log stops at 30k. The 12k–30k columns are paired at $n=3$, and the 32k column alone is unpaired, blank over three seeds against prior over two. Both blank – prior series are printed in full in Table S7.

step (k)	12	14	16	18	20	22	24	26	28	30	32
elbow blank – prior (mrad)	+4.1	+17.8	+2.7	−9.9	+15.1	+30.9	+10.9	−7.1	−5.2	−4.1	−5.4
finger blank – prior (mm)	+26.5	+53.2	+38.8	+52.8	+35.0	+44.6	+30.4	+40.1	+38.6	+67.5	+43.8

Table S7. The supplementary longer-budget probe, blank and prior at $n=3$ seeds, curves parsed from logs and no intervals drawn.

Both series oscillate. On elbow the gap changes sign three times and blank is ahead of the prior at 18k, 26k, 28k, 30k and 32k. On finger the gap is positive at all eleven checkpoints and shows no trend after 14k. We read the two halves symmetrically. The elbow half is used below as reason to doubt the 12k elbow effect. In contrast, the finger half runs the other way. The prior is ahead of blank at every checkpoint out to 32k, the same direction as the 12k prior – blank finger contrast that $n=12$ could not resolve (−27.10). That consistency is not evidence, at $n=3$ and with no intervals; checkpoints within a run are strongly autocorrelated so eleven of them are not eleven observations, and the finger blank series itself swings between 95 and 134 mm over the same stretch. We report it because using this probe only where it costs us, and skipping it where it does not, would be a selection, and the correct conclusion from it on both bodies is the same one, that it supports no claim. These three seeds are seeds 0–2 of the main study, bit-identical through 12k, and their own 12k prior – blank gap is −4.2 mrad against the 12-seed −20.7 (per seed −23.8, +10.6, +0.7) at a blank mean of 57.8 against 70.73, a favourable subsample of the baseline in which the elbow effect had already disappeared *before* the extra budget was spent. They are direct reason to doubt that the 12k elbow effect survives to 30k, and no long-budget run was done with a coach of any schedule.

Four MyoSuite bodies were instantiated and trained on in the course of this study. Two carry the analysis, this subsection reports the third, and the fourth, myoArm, a single truncated blank seed under MyoSuite's own reward, is in Appendix F. Neither the third nor the fourth carries a claim. `results_hand/` holds twelve complete runs on myoHand (39 muscles), {blank, prior} $\times$ six seeds, thirty checkpoints each, 60,000 environment steps, train targets, produced by `launch_hand.ps1` calling the same `run_reach.py`. We detect no improvement on either arm over the budget. Paired within arm over the six seeds, blank at 60k minus blank at 2k is +3.88 mm [−7.10, +14.86] ($p = 0.406$) and prior is −3.51 [−13.81, +6.80] ($p = 0.422$). On the mean of the last five checkpoints minus the mean of the first five, blank is −0.28 [−6.28, +5.71] and prior −3.82 [−12.48, +4.83]. The levels behind those intervals: blank 102.5 mm at 2k and 106.3 at 60k, prior 103.1 and 99.6, both series oscillating flat in the 99–119 mm band for thirty checkpoints. Per §6 an undetected change is a failure to detect, not an absence, so we say that no improvement is detected, and we do not say that neither arm learns.

Paired between arms over the same six seeds, one window cell reaches nominal significance: 12k, prior – blank = +12.49 [+0.99, +23.99], Wilcoxon 0.03125, 6/6 seeds on the sign. It does not survive the window family §3.6's criterion would generate for a body with one condition pair and two endpoints. myoHand's five window cells (EARLY, MID, LATE, the main study's 12k endpoint and this run's own 60k endpoint, a set that keeps the MID and LATE cells §3.6's exclusion criterion would prune, so it is the *unpruned* window family rather than the one that criterion derives) put the 12k cell at Holm rank 1, $p = 0.03832$ against a threshold of $0.05/5 = 0.01000$: it fails, the step-down stops there, and nothing in the family survives. Pruned to $K = 3$ the 12k cell faces 0.0167 and still fails at $p = 0.03832$, so the verdict does not turn on the choice of family. A 30-checkpoint per-checkpoint grid was also computed and returns the same verdict. Two of its 30 cells reach $p <$

0.05 against a chance expectation of 1.5, and Holm over that grid also stops at rank 1. The window family is what §3.6's policy licenses. The per-checkpoint grid is reported for disclosure only. It is not entered into a correction family, which is the exemption §3.6 applies everywhere else. We keep it even on the one body where entering the grid would have removed an unfavourable cell. The window family is used instead precisely so that the exemption is not granted where it helps and imposed where it hurts. The 12k cell's neighbours run +8.34 and -10.62, neither significant, and by 60k the sign has changed again (-6.79 [-22.45, +8.86]). The body is uninformative about the forward-model route. It does not refute it, and no claim in this paper rests on it. The window table is in Appendix C.

M. Literature detail deferred from §2

The offline model-based distinctions and the guidance-hypothesis evidence, in full.

Offline model-based RL. MOREL [8], MOPO [9] and COMBO [10] freeze a pretrained model and then optimize under it, as Plan2Explore's zero-shot mode also does [7], but they train the model on logged interaction data and constrain the optimization, by uncertainty-based pessimism [8,9] or by conservative value regularization [10], COMBO having been built specifically to avoid uncertainty estimation. Only MOREL optimizes purely inside the model. MOPO and COMBO both draw policy-update batches from the logged data as well. We freeze a model learned from reward-free, task-agnostic babbling and use it optimistically. Unlike that family, our prior arm is also handed the environment's analytic reward evaluated at imagined states (§3.5), task information MOPO and COMBO learn from data, their dynamics models predicting reward alongside next state, though MOREL likewise assumes a known reward function (§3 of [8]).

Guidance, augmented feedback, and policy distillation. The guidance hypothesis [25] holds that augmented feedback guides the learner toward the correct response while breeding a dependency on it, so frequent feedback can flatter acquisition relative to retention. We state it that way, and not as the familiar "helps acquisition, hurts retention", because its best-known test supports only the retention half. Winstein and Schmidt [26] report a reliable benefit of reduced feedback at *delayed* no-feedback retention under a faded schedule, while acquisition was not reliably degraded. In their Experiment 2 the reduced-feedback group was significantly *better* during acquisition, but only on the second practice day and only when scored on feedback trials. On Day 1 no group difference appeared even there, and the pooled all-trials analysis found no reliable group effect on either day. Their Experiment 1 detected no reliable group difference in either phase, on a 10-minute test rather than a delayed one. The associated *reversal* effect is meta-analytically weak as well: Marschall et al. [27] report fail-safe N of 2 and 3 against an effect-size threshold of 0.2, and 0 at thresholds of 0.5 and 0.8. The hypothesis's methodological core is the acquisition-versus-retention distinction; a measure taken *while* the aid is present measures acquisition, and only a test *after removal* measures learning. Our coach is a policy-distillation / kickstarting auxiliary, impersonal guidance and never social scaffolding. We separate the two limbs by running the coach at constant weight (the schedule is defined in §3.2 and the constant-versus-annealed contrast is in Appendix D) *and* cutting both aids at a common step (§3.4, results in §4.4).

N. The combination and the frozen-ness experiment

What §4.7 summarises, in full.

priorcoach(const) - prior reproduces the §4.5 crossover with a different condition pair, EARLY elbow -59.30 [-80.25, -38.34] and finger -16.54 [-20.62, -12.46], ENDPOINT elbow -11.53 [-17.80, -5.27] and finger +25.63 [+8.88, +42.38], though non-independently, since it reuses the same runs. Against the coach alone, priorcoach(const) - coach(const) is non-significant at ENDPOINT on both bodies (elbow -0.64 [-4.05, +2.77], finger -6.64 [-14.71, +1.43]): at the endpoint we cannot detect a difference between the combination and the coach alone, which is a null at $n=12$, not a demonstration that combining lands on the coach's outcome. The elbow EARLY cell is significant (-8.74 [-17.09, -0.38]), so the prior is not inert in the presence of a maintained coach on elbow. The matching finger cell is not (-4.82 [-10.18, +0.54]). Full combination tables, including the mixture-midpoint contrasts, are in Appendix D.

Frozen-ness is a design commitment carried by the title, and a dedicated $n=12$ experiment tests it directly. results_frozen12/ holds four arms on myoElbow at the identical 12k budget, twelve matched seeds each on train targets. Two of the four are not new data, run_reach.py being deterministic, so the blank and prior cells re-executed here are byte-identical to their results_2x2_elbow originals, and the prior - blank value of -25.50 mrad is the same number as §4.3's and is not an independent reproduction. What is new is warm and colearn at

n=12. The arms are: prior (this study's frozen babble-learned model), warm (the identical babble checkpoint with freeze=False, fine-tuned online), colearn (a world model trained online from scratch with no babble), and blank. prior – warm isolates freezing cleanly, as the two arms load the same checkpoint and differ only in the freeze flag. warm – colearn does not isolate babbling. As implemented, colearn also uses a different learning rate (1e-3 against 3e-4), re-initializes its model from random weights at every 2k refresh instead of fine-tuning cumulatively, refits its normalizers on real data, holds out 10% for validation, and, because train_world_model reseeds the *global* torch stream that SAC's exploration draws from, receives a periodic RNG reset that warm does not. The re-initialization and the reseeding are each on their own sufficient to generate large seed-to-seed spread, so no causal attribution to babbling is available from this contrast.

Means are prior 40.15 mrad, warm 39.46, colearn 45.66. Freezing is a tight null, with prior – warm = +0.68 [–2.64, +4.01], an interval that at 95% confidence rules out effects beyond about –2.6/+4.0 mrad, set against the prior's own –25.50 mrad effect over blank on the same target set (§4.3), so freezing neither helps nor hurts at a scale that could matter here. This supersedes the n=5 slice in results_myo/, where the same contrast was +1.70 [–9.65, +13.05], an interval that already excluded a headline-sized (≈ 25 mrad) effect but not a quarter-sized one, which is what the n=12 replication narrows. That slice also holds an imag arm (a policy trained inside the frozen prior taking zero real environment steps, 44.55 mrad at n=5), which the n=12 replication does not include. prior – colearn = –5.51 [–19.91, +8.89] and warm – colearn = –6.19 [–19.35, +6.96] both remain unresolved, and the second of those is the contrast this design nominates to isolate babbling, an interval about four times wider than the freezing null beside it, so nothing here shows that babbling improves endpoint error.

What separates the arms is dispersion, not accuracy. What produces that dispersion is not identified. Post hoc, on the apparatus Appendix G declares for §4.8: endpoint SDs are prior 6.27, warm 5.63, colearn 19.77. The online-from-scratch model is between about 1.7 and six times as dispersed as either babble-pretrained arm, SD(colearn)/SD(warm) = 3.51 [1.88, 5.83], Pitman–Morgan $p = 0.00046$; against prior, 3.15 [1.69, 5.32], $p = 0.00077$, while the two babble arms are not distinguishable from each other, SD(prior)/SD(warm) = 1.11 [0.68, 1.76]. However, because colearn differs from warm in five implementation respects, not only in babble-pretraining (above), this study cannot say that the babbling is what buys the reproducibility. The observation is post hoc, on one body and one target set, and carries no numbered claim. Isolating the cause would need colearn re-run with matched learning rate, refresh policy, normalizers and RNG handling.

We therefore describe the frozen prior as *the arm we ran*, since freezing is a defensible implementation choice that this experiment shows to be inconsequential at this scale, and on the accuracy axis neither freezing nor babbling is shown to matter at n=12. The one place the babble-pretrained arms separate from the online-from-scratch one is seed-to-seed dispersion, which is post hoc and carries no numbered claim.

O. The reversal checkpoint by checkpoint

The block §4.5 defers, verbatim.

The reversal, checkpoint by checkpoint. Table S8 gives it.

step	elbow blank	elbow prior	elbow coach	elbow coach – prior	finger blank	finger prior	finger coach	finger coach – prior
2k	172.95	136.24	86.60	–49.64 [–103.16, +3.89] <i>n.s.</i>	143.26	136.20	134.82	–1.38 [–15.76, +13.00] <i>n.s.</i>
4k	124.84	120.45	58.43	–62.02 [–87.43, –36.61]	135.61	133.14	115.39	–17.75 [–25.64, –9.86]
6k	74.58	90.27	50.25	–40.02 [–59.41, –20.63]	136.90	128.80	112.77	–16.03 [–22.87, –9.18]
8k	89.43	60.11	45.52	–14.59 [–28.50, –0.68]	127.95	130.14	111.26	–18.88 [–33.72, –4.03]
10k	81.24	58.12	45.35	–12.77 [–23.78, –1.76]	105.34	107.31	114.95	+7.64 [–3.37, +18.64] <i>n.s.</i>
12k	70.73	50.02	39.13	–10.89 [–19.26, –2.52]	117.60	90.50	122.77	+32.27 [+16.49, +48.05]

Table S8. The reversal checkpoint by checkpoint, all six archived checkpoints, held-out, paired, n=12. Elbow columns are mrad and finger columns are mm. Bold in a mean column marks the two checkpoints at which the numerically lowest of the three is not the constant coach.

Elbow columns are mrad and finger columns are mm. Bold in a mean column marks the two checkpoints at which the numerically lowest of the three is not the constant coach. The finger reversal is directionally consistent but not monotone and is carried by the last 2k; the coach leads at 4k, 6k and 8k (-17.75, -16.03, -18.88), the 10k contrast is +7.64 with a CI containing zero, and most of the +32.27 accrues between 10k and 12k, driven by the prior falling 107.3 → 90.5 while the coach rises 115.0 → 122.8. At 10k model-free blank (105.3 mm) is significantly below the constant coach (paired -9.61 mm [-15.62, -3.60]) but is *not* distinguishable from the prior (-1.97 [-15.23, +11.28]; blank is the numerically lowest of the three in 6 of 12 seeds), and blank itself rises +12.26 mm across 10k→12k, so the stretch carrying the reversal is one in which none of the three curves is behaving like a converged system. We do not localize the crossing more tightly than 8k–12k.

The leading explanation for where the crossing falls is the teacher ceiling of the teachers as captured (§3.4): the clean elbow teacher reaches 27.2 mrad, far better than any student, while the clean finger teacher reaches 169.1 mm, worse than every student condition mean bar one and essentially at the no-op referent, so on finger the student is pulled toward a target worse than what it can find alone. The account is narrower than it looks, because the withdrawal numbers run against the obvious reading of it. If the pull toward 169.1 mm were simply costly, cutting it at 8k should relieve the cost. Instead the kept coach ends at 122.77 and the arm whose coach was cut ends at 169.14, essentially *on* the teacher's own score. We therefore read the finger withdrawal cost (+46.4) as dominated by the optimization shock §4.4 already names rather than by loss of teacher content, and confine the ceiling account to what it does explain, which is why the kept finger coach stops improving and rises over the last 4k (111.3 → 122.8). It does not explain the crossover's main driver either, and the decomposition above puts most of the +32.27 in the prior *falling* 107.3 → 90.5, which a ceiling on the coach says nothing about, and that terminal drop is unexplained. The account also rests on two single-seed terminal checkpoints, and no experiment here varies teacher quality while holding everything else fixed.

References

- [1] Ha & Schmidhuber. *World Models*. arXiv:1803.10122, 2018. (The NeurIPS 2018 camera-ready is a distinct paper: *Recurrent World Models Facilitate Policy Evolution*, arXiv:1809.01999, whose arXiv comments note substantial text overlap with 1803.10122.)
- [2] Janner, Fu, Zhang, Levine. *When to Trust Your Model: Model-Based Policy Optimization*. NeurIPS 2019. arXiv:1906.08253.
- [3] Hafner, Lillicrap, Ba, Norouzi. *Dream to Control: Learning Behaviors by Latent Imagination*. ICLR 2020. arXiv:1912.01603.
- [4] Hafner, Lillicrap, Norouzi, Ba. *Mastering Atari with Discrete World Models*. ICLR 2021. arXiv:2010.02193.
- [5] Hafner, Pasukonis, Ba, Lillicrap. *Mastering diverse control tasks through world models*. Nature 640(8059):647-653, 2025. DOI 10.1038/s41586-025-08744-2. (Preprint: *Mastering Diverse Domains through World Models*, arXiv:2301.04104, 2023.)
- [6] Chua, Calandra, McAllister, Levine. *Deep Reinforcement Learning in a Handful of Trials using Probabilistic Dynamics Models*. NeurIPS 2018. arXiv:1805.12114.
- [7] Sekar, Rybkin, Daniilidis, Abbeel, Hafner, Pathak. *Planning to Explore via Self-Supervised World Models*. ICML 2020. arXiv:2005.05960.
- [8] Kidambi, Rajeswaran, Netrapalli, Joachims. *MOREL: Model-Based Offline Reinforcement Learning*. NeurIPS 2020. arXiv:2005.05951.
- [9] Yu, Thomas, Yu, Ermon, Zou, Levine, Finn, Ma. *MOPO: Model-based Offline Policy Optimization*. NeurIPS 2020. arXiv:2005.13239.
- [10] Yu, Kumar, Rafailov, Rajeswaran, Levine, Finn. *COMBO: Conservative Offline Model-Based Policy Optimization*. NeurIPS 2021. arXiv:2102.08363.
- [11] Kim & Lee. *FreeMusco: Motion-Free Learning of Latent Control for Morphology-Adaptive Locomotion in Musculoskeletal Characters*. SIGGRAPH Asia 2025. arXiv:2511.14205.
- [12] Caggiano, Wang, Durandau, Sartori, Kumar. *MyoSuite: A Contact-Rich Simulation Suite for Musculoskeletal Motor Control*. L4DC 2022. arXiv:2205.13600.
- [13] Berg, Caggiano, Kumar. *SAR: Generalization of Physiological Agility and Dexterity via Synergistic Action Representation*. RSS 2023. arXiv:2307.03716.
- [14] Chiappa, Marin Vargas, Huang, Mathis. *Latent Exploration for Reinforcement Learning*. NeurIPS 2023. arXiv:2305.20065.
- [15] Seo, Lee, James, Abbeel. *Reinforcement Learning with Action-Free Pre-Training from Videos*. ICML 2022. arXiv:2203.13880.
- [16] Wolpert, Ghahramani, Jordan. *An Internal Model for Sensorimotor Integration*. Science 269(5232):1880-1882, 1995.
- [17] Kawato. *Internal Models for Motor Control and Trajectory Planning*. Current Opinion in Neurobiology 9(6):718-727, 1999.
- [18] Schumacher, Haeufle, Büchler, Schmitt, Martius. *DEP-RL: Embodied Exploration for Reinforcement Learning in Overactuated and Musculoskeletal Systems*. ICLR 2023. arXiv:2206.00484.
- [19] Wochner, Schumacher, Martius, Büchler, Schmitt, Haeufle. *Learning with Muscles: Benefits for Data-Efficiency and Robustness in Anthropomorphic Tasks*. CoRL 2022. arXiv:2207.03952.
- [20] Feng, Xu, Liu. *MuscleVAE: Model-Based Controllers of Muscle-Actuated Characters*. SIGGRAPH Asia 2023. arXiv:2312.07340.
- [21] Bongard, Zykov, Lipson. *Resilient Machines Through Continuous Self-Modeling*. Science 314(5802):1118-1121, 2006.
- [22] Rolf, Steil, Gienger. *Goal Babbling Permits Direct Learning of Inverse Kinematics*. IEEE Transactions on Autonomous Mental Development 2(3):216-229, 2010.
- [23] Deisenroth, Rasmussen. *PILCO: A Model-Based and Data-Efficient Approach to Policy Search*. ICML 2011, pp. 465-472.
- [24] Hansen, Su, Wang. *TD-MPC2: Scalable, Robust World Models for Continuous Control*. ICLR 2024. arXiv:2310.16828.
- [25] Salmoni, Schmidt, Walter. *Knowledge of Results and Motor Learning: A Review and Critical Reappraisal*. Psychological Bulletin 95(3):355-386, 1984.
- [26] Winstein, Schmidt. *Reduced Frequency of Knowledge of Results Enhances Motor Skill Learning*. Journal of Experimental Psychology: Learning, Memory, and Cognition 16(4):677-691, 1990.

- [27] Marschall, Bund, Wiemeyer. *Does Frequent Augmented Feedback Really Degrade Learning? A Meta-Analysis*. E-Journal Bewegung und Training 1:74-85, 2007.
- [28] Wolpert, Kawato. *Multiple Paired Forward and Inverse Models for Motor Control*. Neural Networks 11(7-8):1317-1329, 1998.
- [29] Shadmehr, Mussa-Ivaldi. *Adaptive Representation of Dynamics During Learning of a Motor Task*. Journal of Neuroscience 14(5):3208-3224, 1994.